the indexing wizard appears again (as in Figure 2). Equivalent menu options are available under the *History* menu.

Multiple tabs can be opened to have different views of the same code, e.g., classes in one tab, header files in another tab, etc.

We can take a snapshot of any view as an image, using the context menu option 'Save as image'.

Attaching source code from other IDEs

Many times, you may find the source code under existing projects in popular IDEs like Visual Studio, Code Blocks, Qt Creator, etc. Examples are:

- Visual Studio solution for C/C++
- Code Blocks project for C/C++
- Maven/Gradle project for Java

We can also attach source code in the form of clang JSON compilation database. You can generate this as follows for:

- CMake based projects, which define the flag *CMAKE_EXPORT_COMPILE_COMMANDS.*
- Make projects; here, you can run the *make* command in the presence of the external tool Bear. You can build Bear from *https://github.com/rizsotto/Bear* or via package managers like apt in Ubuntu.
- Qt Creator v4.8 (bundled with QtSDK 5.12) onwards comes with an option to generate a compilation database under the *build* menu.

Code editor plugins

Sourcetrail comes with a rich set of plugins to integrate with many popular IDEs/editors like Atom, Clion, Eclipse, Emacs, IntelliJ IDEA, Qt Creator, Sublime, Vim, VS Code and Visual Studio ID. With these plugins you can send the location from the editor to Sourcetrail, to see all the symbols found at that point. Some of these plugins support

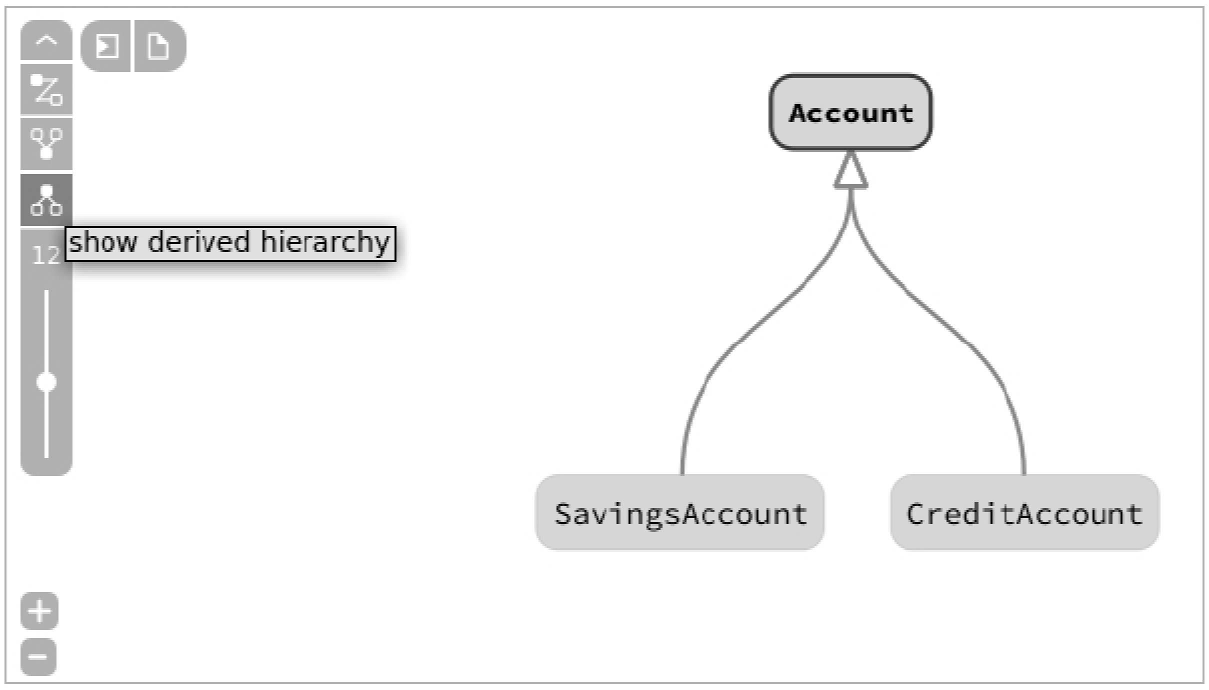

Figure 5: Relationships between classes

generation of the Clang compilation database, and you can attach this with Sourcetrail manually. Also, from the Sourcetrail code pane, you can switch to IDE, with the context menu option 'Show in IDE' for any symbol.

Please refer to *https://www.Sourcetrail.com/documentation/#CodeEditorPlugins* for the IDE/editor of your choice.

Source analysis fest

Here are some interesting libraries in C/C++; do try analysing a few of these with Sourcetrail, depending on your interest and the need to get a better understanding:

- JSON for modern C++: *https://github.com/nlohmann/json*
- iPerf API: *https://iperf.fr/iperf-download.php#source*
- QtSensors API: *https://github.com/qt/qtsensors*
- RTIMU library API: *https://github.com/RPi-Distro/RTIMULib*
- json-c API: *https://github.com/json-c/json-c*
- Point cloud library: *https://github.com/PointCloudLibrary/pcl*
- WiringPi API: *https://github.com/WiringPi/WiringPi*
- mbed driver, RTOS API: *https://github.com/ARMmbed/mbed-os/*
- curl APIs*: https://github.com/curl/curl*
- Paho C library: *https://github.com/eclipse/paho.mqtt.c*

Note: Some of these libraries have rich documentation; you can just compare the official documentation with the Sourcetrail outcome.

You can try some more libraries, as per your domain interest/background, in other supported languages too, such as Java and Python. You can also attach any custom code available with you and visualise it.

This article has given you a glimpse of the features of Sourcetrail, and some initial hands-on pointers. Please explore the tool in depth and gain confidence when dealing with unfamiliar code. END

By: Dr Prithvi Sekhar Pagala and Rajesh Sola

The authors are educators in the Education and Competency Development (ECoDe) team at KPIT Technologies.